# The Time Value of Money

## Basic Concepts

### What is the Time Value of Money?

1. The opportunity to put money to work arises due to the existence of opportunities to invest.
2. The expected rate of return can be used to express the time value of money.

### Why Do We Use the Time Value of Money?

1. To position adjusted cash flows (inflows and outflows) at a single point in time in order to compare and make informed decisions.
2. It applies the concepts of simple interest and compound interest.

## Key Definitions

| Term | Definition |
|---|---|
| **Interest** | Money paid (earned) for the use of money. |
| **Simple Interest** | Interest earned on the original (principal) amount only. |
| **Compound Interest** | Interest earned on previously accumulated interest (interest on interest). |
| **Present Value** | The current value of a future amount of money, or a series of payments, evaluated at a given interest rate. A dollar today is worth more than a dollar tomorrow. |
| **Future Value** | The amount to which an original amount will grow after earning interest. |
| **Discount Rate** | The interest rate used to compute the present value of future cash flows. |
| **Discount Factor** | The present value of a $1 future payment. Expressed as **1/(1+r)^t**. |
| **Annuity** | A series of equal payments or receipts (a stream of cash flows) occurring over a specific number of periods. |
| **Ordinary Annuity** | An equal stream of cash flows occurring at the **end** of each period. |
| **Annuity Due** | An equal stream of cash flows occurring at the **beginning** of each period. |

| Term | Definition |
|---|---|
| **Perpetuity** | An equal stream of cash flows that **never ends**. |
| **Nominal Interest Rate** | The rate of interest quoted for a year. |
| **Effective Annual Interest Rate** | An interest rate annualized using **compound** interest. |
| **Real Interest Rate** | The rate at which the purchasing power of an investment increases. |
| **Annual Percentage Rate (APR)** | An interest rate annualized using **simple** interest. |
| **Amortization Schedule** | A table showing the repayment schedule of interest and principal necessary to pay off a loan by maturity. |

## Core Formulae

1. **Amount of Interest** = Principal Amount × Rate of Interest
2. **Total Amount** = Principal Amount + Amount of Interest
3. **Future Value** = Principal × (1 + r)^t
4. **Present Value** = Future Value × 1/(1 + r)^t

## Topics Covered in Practice

1. Future Value Tables
2. Present Value Tables
3. Annuity
4. Present Value of Future Cash Flows
5. Future Value of Present Cash Flows
6. Multiple Cash Flows

# Practice — The Time Value of Money

## (Activity I)

## Section 1: Simple Interest

Interest earned on the original amount is called **simple interest**.

### Activity 1 — Simple Interest @ 6%

**Table 1:** Simple Interest Calculations

| Year | Balance at Start of Year | Interest Earned @ 6% | Balance at End of Year |
|---|---|---|---|
| 1 | 100 | 100 × 6/100 = 6 | 106 |
| 2 | 100 | 6 | 112 |
| 3 | 100 | 6 | 118 |
| 4 | 100 | 6 | 124 |
| 5 | 100 | 6 | 130 |

### Activity 2 — Simple Interest @ 10% *(Student Practice)*

**Table 2:** Simple Interest Calculations

| Year | Balance at Start of Year | Interest Earned @ 10% | Balance at End of Year |
|---|---|---|---|
| 1 | 1,000 | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |

## Section 2: Compound Interest

Interest earned on previously accumulated interest is called **compound interest**.

### Activity 3 — Compound Interest @ 6%

**Table 3:** Compound Interest Calculations

| Year | Balance at Start of Year | Interest Earned @ 6% | Balance at End of Year |
|---|---|---|---|
| 1 | 100.0000 | 100 × 6/100 = 6.0000 | 106.0000 |
| 2 | 106.0000 | 106 × 6/100 = 6.3600 | 112.3600 |
| 3 | 112.3600 | 112.36 × 6/100 = 6.7416 | 119.1016 |
| 4 | 119.1016 | 119.1016 × 6/100 = 7.1460 | 126.2476 |
| 5 | 126.2476 | 126.2476 × 6/100 = 7.5748 | 133.8226 |

### Activity 4 — Compound Interest @ 10% *(Student Practice)*

**Table 4:** Compound Interest Calculations

| Year | Balance at Start of Year | Interest Earned @ 10% | Balance at End of Year |
|---|---|---|---|
| 1 | 1,000 | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |

## Section 3: Present Value

**Formula:** Present Value = Future Value × 1/(1 + r)^t

### Activity 5 — Present Value Interest Factors @ 1%–10% *(Partial — Student Practice)*

**Table 5:** Present Value Interest Factor = 1/(1 + r)^t

| Years | 1% | 2% | 3% | 4% | 5% | 6% | 7% | 8% | 9% | 10% |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1.00 | | | | | | | | | 1.00 |
| 1 | 0.99 | | | | | | | | | 0.90 |
| 2 | 0.98 | | | | | | | | | 0.81 |
| 3 | 0.97 | | | | | | | | | 0.73 |
| 4 | 0.96 | | | | | | | | | 0.66 |
| 5 | | | | | | | | | | |
| 6 | | | | | | | | | | |
| 7 | | | | | | | | | | |
| 8 | | | | | | | | | | |
| 9 | | | | | | | | | | |
| 10 | | | | | | | | | | |

### Activity 6 — Present Value Interest Factors @ 10%–100% *(Student Practice)*

| Years | 10% | 20% | 30% | 40% | 50% | 60% | 70% | 80% | 90% | 100% |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | | | | | | | | | | |
| 1 | | | | | | | 1/(1+70/100) = | | 1/(1+90/100) = | |
| 2 | | | | | | | | | | |
| 3 | | | | | | | | | | |
| 4 | | | | | | | | | | |
| 5 | | | | | | | | | | |
| 6 | | | | | | | | | | |
| 7 | | | | | | | | | | |
| 8 | | | | | | | | | | |
| 9 | | | | | | | | | | |
| 10 | | | | | | | | | | |

## Activity 7 — Present Value Interest Factors @ 11%–20% *(Student Practice)*

| Years | 11% | 12% | 13% | 14% | 15% | 16% | 17% | 18% | 19% | 20% |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | | | | | | | | | | |
| 1 | | | | | | | | | | |
| 2 | | | | | | | | | | |
| 3 | | | | | | | | | | |
| 4 | | | | | | | | | | |
| 5 | | | | | | | | | | |
| 6 | | | | | | | | | | |
| 7 | | | | | | | | | | |
| 8 | | | | | | | | | | |
| 9 | | | | | | | | | | |
| 10 | | | | | | | | | | |

## Activity 8 — Present Value Calculation @ 15% *(Solved)*

| Years | Cash Flows (Rs.) | P.V.F @ 15% | Present Value (Rs.) |
|---|---|---|---|
| 0 | 10,000 | 1.000 | 10,000 |
| 1 | 15,000 | 0.860 | 12,900 |
| 2 | 20,000 | 0.739 | 14,780 |
| 3 | 10,000 | 0.636 | 6,360 |

| Years | Cash Flows (Rs.) | P.V.F @ 15% | Present Value (Rs.) |
|---|---|---|---|
| 4 | 8,000 | 0.547 | 4,376 |
| 5 | 30,000 | 0.470 | 14,100 |

### Activity 9 — Present Value Calculation @ 10% *(Student Practice)*

| Years | Cash Flows (Rs.) | P.V.F @ 10% | Present Value (Rs.) |
|---|---|---|---|
| 0 | 20,000 | | |
| 1 | 25,000 | | |
| 2 | 20,000 | | |
| 3 | 30,000 | | |
| 4 | 80,000 | | |
| 5 | 90,000 | | |

# Section 4: Future Value

**Formula:** Future Value = Principal × (1 + r)^t

### Activity 10 — Future Value Interest Factors @ 1%–10% *(Partial — Student Practice)*

| Years | 1% | 2% | 3% | 4% | 5% | 6% | 7% | 8% | 9% | 10% |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | | | | | | | | | |
| 1 | 1.01 | | | | 1.05 | | | | | 1.10 |
| 2 | 1.02 | | | | | | | | | 1.21 |
| 3 | 1.03 | | | | | | | | | 1.33 |
| 4 | | | | | | | | | | 1.46 |
| 5 | | | | | | | | | | |
| 6 | | | | | | | | | | |
| 7 | | | | | | | | | | |
| 8 | | | | | | | | | | |
| 9 | | | | | | | | | | |
| 10 | | | | | | | | | | |

### Activity 11 — Future Value Interest Factors @ 10%–100% *(Student Practice)*

| Years | 10% | 20% | 30% | 40% | 50% | 60% | 70% | 80% | 90% | 100% |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | | | | | | | | | | |
| 1 | | | | | | | | | | |
| 2 | | | | | | | | | | |
| 3 | | | | | | | | | | |
| 4 | | | | | | | | | | |
| 5 | | | | | | | | | | |
| 6 | | | | | | | | | | |
| 7 | | | | | | | | | | |
| 8 | | | | | | | | | | |
| 9 | | | | | | | | | | |
| 10 | | | | | | | | | | |

### Activity 12 — Future Value Interest Factors @ 11%–20% *(Student Practice)*

| Years | 11% | 12% | 13% | 14% | 15% | 16% | 17% | 18% | 19% | 20% |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | | | | | | | | | | |
| 1 | | | | | | | | | | |
| 2 | | | | | | | | | | |
| 3 | | | | | | | | | | |
| 4 | | | | | | | | | | |
| 5 | | | | | | | | | | |
| 6 | | | | | | | | | | |
| 7 | | | | | | | | | | |
| 8 | | | | | | | | | | |
| 9 | | | | | | | | | | |
| 10 | | | | | | | | | | |

### Activity 13 — Future Value Calculation @ 15% *(Student Practice)*

| Years | Cash Flows (Rs.) | F.V.F @ 15% | Future Value (Rs.) |
|---|---|---|---|
| 0 | 10,000 | | |
| 1 | 15,000 | | |
| 2 | 20,000 | | |
| 3 | 10,000 | | |

| Years | Cash Flows (Rs.) | F.V.F @ 15% | Future Value (Rs.) |
|---|---|---|---|
| 4 | 8,000 | | |
| 5 | 30,000 | | |

### Activity 14 — Future Value Calculation @ 25% *(Student Practice)*

| Years | Cash Flows (Rs.) | F.V.F @ 25% | Future Value (Rs.) |
|---|---|---|---|
| 0 | 100,000 | | |
| 1 | 150,000 | | |
| 2 | 200,000 | | |
| 3 | 100,000 | | |
| 4 | 80,000 | | |
| 5 | 300,000 | | |

# Section 5: Solved Examples

**Example 1:** Rs. 100 deposited at 6% interest per year.

*Required:*

1. How much interest would be earned after Year 1?
2. What is the total deposit at the end of Year 1?
3. What is the interest earned in Year 2?
4. What is the total deposit at the end of Year 2?
5. What is the interest earned in Year 3?
6. What is the total deposit at the end of Year 3?

**Example 2:** Rs. 500 deposited at 10% interest per year for 20 years.

*Required:*

1. What is the total deposit at the end of 20 years?

**Example 3:** Rs. 500,000 deposited at 17% interest per year for 15 years.

*Required:*

1. What is the total deposit at the end of 15 years?

**Example 4:** Rs. 2,500,000 deposited at 13% interest per year for 12 years.

*Required:*

1. What is the total deposit at the end of 12 years?

# Amortization Schedule

## Basic Concepts

An **amortization schedule** is a table showing the repayment schedule of interest and principal necessary to pay off a loan by maturity.

Each installment payment covers two components:

- **Interest Amount** = Beginning Balance × Interest Rate
- **Principal Amount** = Loan Installment − Interest Amount
- **Outstanding Balance** = Beginning Balance − Principal Amount

## Key Formulae

| Scenario | Formula |
|---|---|
| **Installment paid at end of year** | Equated Annual Installment = Loan Amount ÷ PVIFA |
| **Installment paid at beginning of year (t = 0)** | Equated Annual Installment = Loan Amount ÷ (PVIFA + 1) |

## Solved Examples

**Example 1 — End of Year Payments**

Loan: $40 | Rate: 20% | Term: 5 years

Installment = $40 / PVIFA (20%, 5 yrs) = $40 / 2.990 = **$13.38**

**Example 2 — Beginning of Year Payments**

Loan: $40 | Rate: 20% | Term: 5 years (installment due at beginning)

Installment = $40 / (PVIFA (20%, 4 yrs) + 1) = $40 / (2.588 + 1) = **$11.14**

# Practice Activities

### Activity 1 — $35,000 Loan @ 13% for 5 Years *(End of Year Payments — Student Practice)*

Installment = $35,000 / PVIFA (13%, 5 yrs)

| Year | Loan Installment | Beginning Balance | Interest @ 13% | Principal Amount | Outstanding Balance |
|---|---:|---:|---:|---:|---:|
| 0 | — | — | — | — | 35,000 |
| 1 | | | | | |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |

### Activity 2 — $75,000 Loan @ 12% for 6 Years *(End of Year Payments — Partially Solved)*

Installment = $75,000 / PVIFA (12%, 6 yrs) = $75,000 / 4.111 = **$18,244**

| Year | Loan Installment | Beginning Balance | Interest @ 12% | Principal Amount | Outstanding Balance |
|---|---:|---:|---:|---:|---:|
| 1 | 18,244 | 75,000 | 75,000 × 0.12 = | 18,244 − 9,000 = | 75,000 − 9,244 = |

| Year | Loan Installment | Beginning Balance | Interest @ 12% | Principal Amount | Outstanding Balance |
|---|---|---|---|---|---|
| | | | 9,000 | 9,244 | 65,756 |
| 2 | 18,244 | 65,756 | | | |
| 3 | 18,244 | | | | |
| 4 | 18,244 | | | | |
| 5 | 18,244 | | | | |
| 6 | 18,244 | | | | |

### Activity 3 — $155,000 Loan @ 15% for 8 Years *(End of Year Payments — Partially Solved)*

Installment = $155,000 / PVIFA (15%, 8 yrs) = $155,000 / 4.487 = **$34,544**

| Year | Loan Installment | Beginning Balance | Interest @ 15% | Principal Amount | Outstanding Balance |
|---|---|---|---|---|---|
| 1 | 34,544 | 1,55,000 | 155,000 × 0.15 = 23,250 | 34,544 − 23,250 = 11,294 | 155,000 − 11,294 = 1,43,706 |
| 2 | 34,544 | 1,43,706 | | | |
| 3 | 34,544 | | | | |
| 4 | 34,544 | | | | |
| 5 | 34,544 | | | | |
| 6 | 34,544 | | | | |
| 7 | 34,544 | | | | |
| 8 | 34,544 | | | | |

### Activity 4 — $250,000 Loan @ 25% for 9 Years *(Beginning of Year Payments — Partially Solved)*

Installment = $250,000 / (PVIFA (25%, 8 yrs) + 1) = $250,000 / (3.329 + 1) = **$57,763**

**Note:** At Year 0, no interest accrues — the full installment reduces principal directly.

| Year | Loan Installment | Beginning Balance | Interest @ 25% | Principal Amount | Outstanding Balance |
|---|---|---|---|---|---|
| 0 | 57,763 | 2,50,000 | 0 | 57,763 | 2,50,000 − 57,763 = 1,92,237 |
| 1 | 57,763 | 1,92,237 | | | |

| Year | Loan Installment | Beginning Balance | Interest @ 25% | Principal Amount | Outstanding Balance |
|---|---|---|---|---|---|
| 2 | 57,763 | | | | |
| 3 | 57,763 | | | | |
| 4 | 57,763 | | | | |
| 5 | 57,763 | | | | |
| 6 | 57,763 | | | | |
| 7 | 57,763 | | | | |
| 8 | 57,763 | | | | |

### Activity 5 — $500,000 Loan @ 15% for 4 Years *(Beginning of Year Payments — Partially Solved)*

Installment = $500,000 / (PVIFA (15%, 3 yrs) + 1) = $500,000 / (2.283 + 1) = **$152,300**

| Year | Loan Installment | Beginning Balance | Interest @ 15% | Principal Amount | Outstanding Balance |
|---|---|---|---|---|---|
| 0 | 1,52,300 | 5,00,000 | 0 | 1,52,300 | 5,00,000 − 1,52,300 = 3,47,700 |
| 1 | 1,52,300 | 3,47,700 | | | |
| 2 | 1,52,300 | | | | |
| 3 | 1,52,300 | | | | |
| 4 | 1,52,300 | | | | |

### Activity 6 — $100,000 Loan @ 10% for 7 Years *(Beginning of Year Payments — Student Practice)*

Installment = $100,000 / (PVIFA (10%, 6 yrs) + 1)

| Year | Loan Installment | Beginning Balance | Interest @ 10% | Principal Amount | Outstanding Balance |
|---|---|---|---|---|---|
| 0 | | 1,00,000 | 0 | | |
| 1 | | | | | |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |

| Year | Loan Installment | Beginning Balance | Interest @ 10% | Principal Amount | Outstanding Balance |
|---|---|---|---|---|---|
| 6 | | | | | |
| 7 | | | | | |

# Financial Ratio Analysis

## Part I: Short-Term Liquidity Ratios

### Conceptual Overview

Liquidity ratios measure a firm's ability to meet its current obligations. They assess a business's capacity to convert its assets to cash and pay off obligations without significant difficulty (i.e., without delay or loss of value). These ratios are particularly useful for suppliers, employees, banks, and other stakeholders.

### Key Liquidity Ratios & Formulae

| Ratio | Formula |
|---|---|
| **Current Ratio** | Current Assets / Current Liabilities |
| **Quick Ratio** *(Acid-Test Ratio)* | Quick Assets / Current Liabilities |
| **Cash Ratio** | (Cash + Cash Equivalents + Marketable Securities) / Current Liabilities |
| **Working Capital** | Current Assets |
| **Net Working Capital** | Current Assets − Current Liabilities |
| **Working Capital Ratio** | Current Assets / Current Liabilities *(same as Current Ratio)* |

**Note:** Quick Assets = Current Assets − (Inventory + Prepaid Expenses)

### Practice Questions — Liquidity Ratios

**Question 1:** Beta Co. has a loan covenant requiring it to maintain a current ratio of 1.5 or better. As Beta Co. approaches year-end, current assets are $20 million ($1 million cash, $9 million accounts receivable, $10 million inventory) and current liabilities are $13.5 million.

*Required:*

1. Calculate Beta's current ratio.
2. Calculate Beta's quick ratio.
3. Calculate Beta's cash ratio.

**Question 2:** The following data are adapted from a recent annual report of G.F.C. Limited, a leading fans manufacturer.

| **Balance Sheet Data** | **2002** | **2001** |
|---|---|---|
| Quick assets | 1,304.7 | 815.2 |
| Current assets | 5,166.5 | 4,393.9 |
| Current liabilities | 2,955.2 | 3,011.6 |

*Required:*

(a) Compute the following for 2002 and 2001 (round to one decimal place):

1. Working capital.
2. Net working capital.
3. Current ratio.
4. Quick ratio.

(b) Comment on the trends in the liquidity measures and state whether G.F.C. Limited appears to be solvent at the end of 2002.

**Question 3:** The following financial data was taken from the annual statements of Smith Corporation.

| | **2009** | **2010** | **2011** |
|---|---|---|---|
| Current assets | $450,000 | $400,000 | $500,000 |
| Current liabilities | 390,000 | 300,000 | 340,000 |
| Inventory | 280,000 | 200,000 | 250,000 |

*Required:* Based on the data above, calculate the following for 2010 and 2011:

1. Working capital.
2. Net working capital.
3. Current ratio.
4. Quick ratio.
5. Comment on the trends in short-term liquidity from 2009 to 2011.

**Question 4:**

**FOXX COMPANY — Balance Sheet** *December 31, Year 2*

| Assets | | Liabilities & Equity | |
|---|---|---|---|
| **Current assets:** | | **Current liabilities:** | $250,000 |
| Cash | $75,000 | **Noncurrent liabilities:** | $150,000 |
| Accounts receivable | $75,000 | **Total equity:** | $100,000 |
| Inventory | $50,000 | | |
| Noncurrent assets | $300,000 | | |
| **Total assets** | **$500,000** | **Total liabilities & equity** | **$500,000** |

*Required:*

1. Current ratio.
2. Quick ratio.
3. Cash ratio.
4. Working capital.
5. Net working capital.

**Question 5:** A partial balance sheet and income statement for King Corporation follow.

**Assets — Current:**

| Item | Amount |
|---|---|
| Cash | $33,493 |
| Marketable securities | $215,147 |
| Trade receivables | $255,000 |
| Inventories | $523,000 |
| Prepaid expenses | $26,180 |
| **Total current assets** | **$1,052,820** |

**Liabilities — Current:**

| Item | Amount |
|---|---|
| Trade accounts payable | $103,689 |
| Notes payable | $210,381 |

| Item | Amount |
|---|---|
| Accrued expenses and other liabilities | $120,602 |
| Income taxes payable | $3,120 |
| Current maturities of long-term debt | $22,050 |
| **Total current liabilities** | **$459,842** |

*Required:*

1. Working capital.
2. Net working capital.
3. Current ratio.
4. Acid-test ratio / Quick ratio.
5. Cash ratio.

# Part II: Activity Ratios

### Conceptual Overview

Activity ratios measure how efficiently a firm manages its assets and converts them into sales or cash. The key activity ratios are summarised below.

### Key Activity Ratios & Formulae

| Ratio | Formula | Unit |
|---|---|---|
| **Inventory Turnover** | Cost of Goods Sold / Average Inventory | Times |
| **Inventory Turnover in Days** | Average Inventory / Cost of Goods Sold × 365 | Days |
| **Days' Sales in Inventory** | Ending Inventory / Cost of Goods Sold × 365 | Days |
| **Accounts Receivable Turnover** | Net Sales / Average Gross Receivables | Times |
| **Accounts Receivable Turnover in Days** | Average Gross Receivables / Net Sales × 365 | Days |
| **Days' Sales in Receivables** | Ending Gross Receivables / Net Sales × 365 | Days |
| **Accounts Payable Turnover** | Net Credit Purchases / Average Accounts Payable | Times |
| **Accounts Payable Turnover in Days** | Average Accounts Payable / Net Credit Purchases × 365 | Days |

| Ratio | Formula | Unit |
|---|---|---|
| **Days' Payable Turnover** | Ending Gross Payables / Net Credit Purchases × 365 | Days |
| **Operating Cycle** | AR Turnover in Days + Inventory Turnover in Days | Days |
| **Cash Cycle** | AR Turnover in Days + Inventory Turnover in Days − AP Turnover in Days | Days |

**Average Inventory** = (Opening Inventory + Closing Inventory) / 2

## Practice Questions — Activity Ratios

**Question 1:** The following average receivable and sales data for Shahtaj Sugar Mills are given below.

| Year | Sales | Average Accounts Receivable |
|---|---|---|
| 2002 | 17,000 | 1,700 |
| 2003 | 14,580 | 1,620 |
| 2004 | 9,600 | 1,600 |
| 2005 | 9,000 | 1,800 |

*Required:* Calculate receivable turnover and receivable turnover in days, and provide your expert opinion on:

1. Performance of the collection department from 2002 to 2005.
2. Changes in collection trends from 2002 to 2005.

**Question 2:** The following inventory, receivable, and sales data for Ashraf Company are given below.

| Item | End of Year | Beginning of Year |
|---|---|---|
| Net sales | 31,50,000 | — |
| Gross accounts receivable | 1,80,000 | 1,60,000 |
| Inventory | 4,80,000 | 3,90,000 |
| Cost of goods sold | 22,50,000 | — |

*Required:*

1. Average accounts receivable.
2. Accounts receivable turnover.
3. Accounts receivable turnover in days.
4. Days' receivable turnover.
5. Average inventory.
6. Inventory turnover.
7. Inventory turnover in days.
8. Number of days' sales in inventory.
9. Operating cycle.

**Question 3:** The following data has been provided for Shahzan International Limited for 1986–1987.

| Item | 1987 | 1986 |
|---|---|---|
| Cost of goods sold | 786,523 | 700,263 |
| Opening inventory (Jan 1) | 173,999 | 167,286 |
| Closing inventory (Dec 31) | 195,512 | 173,999 |
| Total inventory | 369,511 | 341,285 |
| Net sales | 1,086,944 | 988,417 |
| Opening working capital (Jan 1) | 241,952 | 217,848 |
| Closing working capital (Dec 31) | 266,586 | 241,952 |
| Total working capital | 508,538 | 459,800 |

*Required:* Calculate the following and comment on the activity ratios of the company.

1. Average inventory.
2. Inventory turnover.
3. Inventory turnover in days.
4. Number of days' sales in inventory.
5. Working capital turnover.
6. Working capital turnover in days.
7. Number of days' sales invested in working capital.

**Question 4:** From the accounts of Arnold Ltd. for the year ended 31 December 2001.

| Item | Amount |
|---|---|
| Sales | 125,000 |
| Opening stock | 20,000 |
| Purchases on credit | 110,000 |

| Item | Amount |
|---|---|
| Gross profit | 25,000 |
| Expenses | 15,000 |
| Fixed assets | 76,000 |
| Current assets (including debtors $25,000) | 66,000 |
| Liquid assets | 36,000 |
| Current liabilities (creditors only) | 22,000 |
| Ordinary shares | 80,000 |
| 8% Cumulative preference shares | 30,000 |
| Capital employed | 120,000 |

*Required:*

1. (a) Cost of goods sold. (b) Closing stock.
2. Period of credit given to debtors (accounts receivable turnover in days).
3. Accounts payable turnover.
4. Period of credit received from creditors (accounts payable turnover in days).
5. Rate of stock turnover (inventory turnover).
6. Inventory turnover in days.

**Question 5:** Selected data for two years ending December 31 for Suleman Ltd.

| Item | Year 1 | Year 2 |
|---|---|---|
| Net sales | 10,50,000 | 10,00,000 |
| Average total assets | 2,30,000 | 2,00,000 |

*Required:* Calculate the total asset turnover from Year 1 to Year 2.

**Question 6:**

| Year | Cost of Goods Sold | Average Stock |
|---|---|---|
| 2002 | 15,000 | 1,000 |
| 2003 | 9,600 | 1,000 |
| 2004 | 14,580 | 1,620 |
| 2005 | 17,000 | 1,740 |

*Required:*

1. What is meant by inventory turnover? How can an entity suffer from poor inventory management?
2. Calculate inventory turnover and inventory turnover in days, and comment on the trends.

**Question 7:** The following financial data was taken from the annual statements of Smith Corporation.

| Item | 2009 | 2010 | 2011 |
|---|---|---|---|
| Sales | 1,450,000 | 1,500,000 | 1,400,000 |
| Cost of goods sold | 1,180,000 | 1,020,000 | 1,120,000 |
| Inventory | 280,000 | 200,000 | 250,000 |
| Accounts receivable | 120,000 | 110,000 | 105,000 |

*Required:* Based on the data above, calculate the following for 2010 and 2011:

1. Accounts receivable turnover.
2. Accounts receivable turnover in days.
3. Inventory turnover.
4. Inventory turnover in days.
5. Operating cycle.

**Question 8:** From the following annual accounts of New Horizontal Limited.

**Extracts from Balance Sheet — 31 December 2007**

| Item | Amount |
|---|---|
| Stocks | 310,000 |
| Debtors | 770,000 |

**Extracts from Profit & Loss Account**

| Item | Amount |
|---|---|
| Sales for the year | 31,00,000 |
| Gross profit | 17,25,000 |
| Expenses | 8,05,000 |
| Depreciation | 2,50,000 |

*Required:* Calculate the following ratios and comment on the results.

1. Debtors turnover.
2. Debtors collection period.
3. Cost of goods sold.
4. Stock turnover.
5. Stock turnover in days.
6. Operating cycle.

**Question 9:** Hawk Company — selected accounts for December 31, 2011 and 2010.

| Item | 2011 | 2010 |
|---|---|---|
| Net sales | $1,180,178 | $2,200,000 |
| Receivables (net) — beginning of year *(allowance: 2011 — $12,300; 2010 — $7,180)* | 240,360 | 230,180 |
| Receivables (net) — end of year *(allowance: 2011 — $11,180; 2010 — $12,300)* | 220,385 | 240,360 |

*Required:*

1. Gross accounts receivable at the beginning of the year for 2011 and 2010.
2. Gross accounts receivable at the end of the year for 2011 and 2010.
3. Average gross receivables for 2011 and 2010.
4. Accounts receivable turnover for 2011 and 2010.
5. Accounts receivable turnover in days for 2011 and 2010.
6. Days' sales in receivables for 2011 and 2010.
7. Accounts receivable turnover using year-beginning gross receivables for 2011 and 2010; comment on the liquidity of Hawk Company's receivables.

**Question 10:** Haward Company — selected accounts for December 31, 2007 and 2006.

| Item | 2007 | 2006 |
|---|---|---|
| Net sales | $2,180,178 | $2,300,000 |
| Receivables (net) — beginning of year *(allowance: 2007 — $12,300; 2006 — $7,180)* | 640,360 | 530,180 |
| Receivables (net) — end of year *(allowance: 2007 — $11,180; 2006 — $12,300)* | 620,385 | 540,360 |

*Required:*

1. Gross accounts receivable at the beginning of the year for 2007 and 2006.
2. Gross accounts receivable at the end of the year for 2007 and 2006.

3. Average gross receivables for 2007 and 2006.
4. Accounts receivable turnover for 2007 and 2006.
5. Accounts receivable turnover in days for 2007 and 2006.
6. Days' sales in receivables for 2007 and 2006.
7. Accounts receivable turnover using year-beginning gross receivables for 2007 and 2006; comment on the liquidity of Haward Company's receivables.

**Question 11:**

**KING CORPORATION — Partial Balance Sheet** *December 31, 2017*

| Item | Amount |
|---|---|
| Trade receivables | $255,000 |
| Inventories | $523,000 |
| Trade accounts payable | $103,689 |
| Notes payable | $210,381 |
| Accrued expenses and other liabilities | $120,602 |

**KING CORPORATION — Partial Income Statement** *For Year Ended December 31, 2017*

| Item | Amount |
|---|---|
| Net sales | $3,050,600 |
| Miscellaneous income | $45,060 |
| **Total revenue** | **$3,095,660** |
| Cost of sales | $2,185,100 |
| Selling, general & administrative expenses | $350,265 |
| Interest expense | $45,600 |
| Income taxes | $300,000 |
| **Total costs & expenses** | **$2,880,965** |
| **Net income** | **$214,695** |

**Note:** Trade receivables at December 31, 2016 were $280,000. Inventory at December 31, 2016 was $565,000.

*Required:*

1. Accounts receivable turnover.
2. Accounts receivable turnover in days.
3. Days' accounts receivable turnover.
4. Inventory turnover.

5. Inventory turnover in days.
6. Number of days' sales in inventory.
7. Operating cycle.

# Cost of Capital

## Part I: Basic Concepts

### Q1(a). What are Capital Components and Their Costs?

The four **capital components** are:

1. Debt
2. Preferred stock
3. Retained earnings
4. Common equity / owner's equity / share capital

The cost of each component is:

1. Cost of debt and after-tax cost of debt
2. Cost of preferred stock
3. Cost of common equity (retained earnings)
4. Cost of external equity (new issue)

### Q1(b). Define and Explain the Following Terms

| Term | Definition |
|---|---|
| **Capital Components** | The types of capital used by a firm: debt, preferred stock, retained earnings, and common equity. |
| **Before-tax cost of debt, r_d** | The interest rate the firm must pay on new debt, before accounting for tax effects. |
| **After-tax cost of debt, r_d(1 – T)** | The relevant cost of new debt, adjusted for the tax deductibility of interest. |
| **Cost of preferred stock, r_p** | The rate of return required by preferred shareholders. |
| **Cost of retained earnings, r_s** | The required rate of return by common stockholders on the firm's existing equity. |
| **Cost of new common stock, r_e** | The cost of external equity, higher than retained earnings due to flotation costs. |
| **Weighted Average Cost of Capital (WACC)** | A weighted average of the component costs of debt, preferred stock, and common equity. |

| Term | Definition |
|---|---|
| **Flotation cost** | Costs associated with issuing new securities (underwriting, legal, listing, and printing fees). |
| **Flotation cost adjustment** | The upward adjustment made to the cost of equity to account for flotation costs when issuing new shares. |

## Q1(c). Pecking Order Theory of Capital Structure

1. The **Pecking Order Theory** holds that firms prefer to finance investments first with retained earnings, then with debt, and lastly with new equity — following a hierarchy based on the cost and information asymmetry of each source.
2. It is useful in selecting the **optimal capital structure** because it provides a practical, observable framework for how firms prioritize financing, minimizing adverse signalling to the market.
3. It **impacts the cost of capital** by suggesting that internally generated funds (retained earnings) carry the lowest cost, while new equity issuance is most expensive due to flotation costs and negative market signals.

## Q2. What is the Cost of Debt?

| Type | Definition |
|---|---|
| **Before-tax cost of debt** | The interest rate the firm must pay on new debt. |
| **After-tax cost of debt** | The relevant cost of new debt, taking into account the tax deductibility of interest: **$r_d(1 - T)$**. |

## Q3. What is the Cost of Preferred Stock?

The rate of return an investor requires on the firm's preferred stock. It is calculated as the preferred dividend ($D_p$) divided by the current price ($P_p$) of the preferred stock.

**$r_p = D_{ps} / P_{ps}$**

Preferred dividends are **not tax-deductible**, so the before-tax and after-tax costs of preferred stock are equal.

## Q4. What is the Cost of Retained Earnings?

The required rate of return by stockholders on a firm's common stock. Since retained earnings belong to existing shareholders, the opportunity cost of using them is equivalent to what those shareholders could earn elsewhere.

## Q5. What is the Cost of New Common Stock?

1. The cost of external equity is based on the cost of retained earnings but **increased for flotation costs**.
2. Equity raised by issuing new stock therefore has a **higher cost** than retained earnings.
3. After the startup stage, firms typically obtain new equity through retained earnings rather than new stock issuances.

## Q6. What Methods are Used to Calculate the Cost of Equity?

**1. Dividend Discount Model (DDM)**

$r_e = D_1/P_0 + g$

**2. Capital Asset Pricing Model (CAPM)**

$r_e = R_f + (R_m - R_f) \times \beta$

*Difficulties with CAPM:*

- Controversy over which risk-free rate to use (long-term vs. short-term)
- Expected beta is difficult to estimate
- Market risk premium is difficult to determine accurately

**3. Bond Yield Plus Risk Premium Approach**

Cost of equity = Bond yield + Risk premium

**4. Dividend Yield Plus Growth Rate (Discounted Cash Flow Approach)**

$r_e = D_1/(1+r)^1 + D_1/(1+r)^2 + \cdots$

**5. Averaging the Methods** If management is highly confident in one method, it may use that method alone. Otherwise, an average of the applicable methods is recommended.

## Q7. What is the Weighted Average Cost of Capital (WACC)?

A weighted average of the component costs of debt, preferred stock, and common equity, where the weights reflect the proportion of each component in the firm's capital structure.

# Part II: Cost of Preferred Stock — Practice Problems

**Formula:** $r_p = D_{ps} / (P_{ps} \times (1 - F))$ Where: $D_{ps}$ = Preferred Dividend, $P_{ps}$ = Stock Price, F = Flotation Cost (%)

**Question 1:** Preferred stock trades at Rs. 20/share; annual dividend Rs. 6/share. Ignore flotation costs.

*Required:* What is the firm's cost of preferred stock?

**Question 2:** Tunney Industries can issue perpetual preferred stock at $47.50/share paying a constant annual dividend of $3.80/share.

*Required:* What is the company's cost of preferred stock, $R_p$?

**Question 3:** Preferred stock trades at Rs. 25/share; annual dividend Rs. 7/share. Ignore flotation costs.

*Required:* What is the firm's cost of preferred stock?

**Question 4:** Preferred stock trades at Rs. 17/share; annual dividend Rs. 4/share. Ignore flotation costs.

*Required:* What is the firm's cost of preferred stock?

**Question 5:** Preferred stock trades at Rs. 30/share; annual dividend Rs. 6/share. Ignore flotation costs.

*Required:* What is the firm's cost of preferred stock?

**Question 6:** Preferred stock trades at Rs. 100/share; annual dividend Rs. 14/share; flotation cost Rs. 10.

*Required:*

1. What is the firm's cost of preferred stock?
2. What is the firm's cost of preferred stock after the flotation cost?

# Part III: Cost of Debt — Practice Problems

**Formula:** After-tax cost of debt = r_d × (1 − T)

**Question 1:** Calculate the after-tax cost of debt under each of the following conditions:

| Condition | Interest Rate | Tax Rate |
|---|---|---|
| a | 13% | 0% |
| b | 13% | 20% |
| c | 13% | 35% |

**Question 2:** Calculate the after-tax cost of debt under each of the following conditions:

| Condition | Interest Rate | Tax Rate |
|---|---|---|
| a | 16% | 10% |
| b | 16% | 24% |
| c | 16% | 35% |

**Question 3:** Calculate the after-tax cost of debt under each of the following conditions:

| Condition | Interest Rate | Tax Rate |
|---|---|---|
| a | 9% | 14% |
| b | 11% | 25% |
| c | 12% | 37% |

**Question 4:** LL Incorporated's currently outstanding 11% coupon bonds have a yield to maturity of 8%. LL believes it could issue new bonds at par with a similar YTM. Marginal tax rate is 35%.

*Required:* What is LL's after-tax cost of debt?

**Question 5:** LL Incorporated's currently outstanding 9.2% coupon bonds have a yield to maturity of 6.9%. Marginal tax rate is 27%.

*Required:* What is LL's after-tax cost of debt?

**Question 6:** A company's 6% coupon rate, semiannual payment, $1,000 par value bond matures in 30 years and sells at $515.16. Federal-plus-state tax rate is 40%.

*Required:* What is the firm's after-tax component cost of debt for WACC purposes? *(Hint: Base your answer on the nominal rate.)*

**Question 7:** A company's 8.7% coupon rate, semiannual payment, $1,200 par value bond matures in 14 years and sells at $734.50. Tax rate is 29%.

*Required:* What is the firm's after-tax component cost of debt for WACC purposes?

**Question 8:** A company will issue new 20-year debt with a par value of $1,000 and a coupon rate of 9%, paid annually. Tax rate is 40%. Flotation cost is 2% of issue proceeds.

*Required:* What is the after-tax cost of debt?

# Part IV: Cost of Equity — Practice Problems

**Formula (DDM):** $r\_e = D_1/P_0 + g$ **Formula (with flotation):** $r\_e = D_1 / [P_0(1 - F)] + g$

**Question 1:** Allied stock sells for Rs. 23.06; expected dividend Rs. 1.25; expected growth rate 8.3%.

*Required:* What is the required rate of return on equity?

**Question 2:** Allied stock sells for Rs. 22.06; expected dividend Rs. 1.25; expected growth rate 7.3%.

*Required:* What is the required rate of return on equity?

**Question 3:** Carpetto Technologies' stock sells for $23.00/share; last dividend was $2.00; expected growth rate 7% per year.

*Required:* What is the cost of common equity?

**Question 4:** Allied stock sells for Rs. 23.06; expected dividend Rs. 1.25; expected growth rate 9.3%.

*Required:* What is the required rate of return on equity?

**Question 5:** Allied stock sells for Rs. 25.00; expected dividend Rs. 1.25; expected growth rate 8.3%.

*Required:* What is the required rate of return on equity?

**Question 6:** Allied stock sells for Rs. 30; expected dividend Rs. 3.25; expected growth rate 8.3%.

*Required:* What is the required rate of return on equity?

**Question 7:** Javits & Sons' common stock trades at $30.00/share. Expected annual dividend is $3.00/share; constant growth rate is 5% per year.

*Required:*

1. What is the cost of common equity if all equity comes from retained earnings?
2. If the company issues new stock with a 10% flotation cost, what is the cost of equity from new stock?

**Question 8:** The Evanec Company's next expected dividend $D_1$ is \$3.18; growth rate is 6%; stock currently sells for \$36.00. New stock can be sold at a net of \$32.40/share.

*Required:*

1. What is the present cost of equity?
2. What is the amount of the flotation cost?
3. What is Evanec's percentage flotation cost, F?
4. What is Evanec's cost of new common stock, $R_e$?

# Weighted Average Cost of Capital (WACC)

## Conceptual Overview

The **Weighted Average Cost of Capital (WACC)** is a weighted average of the component costs of debt, preferred stock, and common equity. The target proportions of each component — along with their respective costs — are used to calculate WACC.

$$WACC = w_d \times r_d(1-T) + w_p \times r_p + w_e \times r_e$$

Where:

- $w_d, w_p, w_e$ = weights of debt, preferred stock, and equity respectively
- $r_d(1-T)$ = after-tax cost of debt
- $r_p$ = cost of preferred stock
- $r_e$ = cost of common equity

## Practice Problems

### Section A: WACC from Given Capital Structure Data

**Question 1(a):** Calculate WACC for Usman Corporation. *(Solved)*

| Capital Component | Cost % | Total Financing | Proportion | Weighted Cost |
|---|---|---|---|---|
| Debt | 6.6% | 30,000 | 30% | 0.0198 |
| Preferred Stock | 10.2% | 10,000 | 10% | 0.0102 |
| Common Stock | 14.0% | 60,000 | 60% | 0.0840 |
| **Total** | | **100,000** | **WACC** | **0.114 (11.4%)** |

**Question 1(b):** Calculate WACC for Usman Corporation. *(Student Practice)*

| Capital Component | Cost % | Total Financing | Proportion | Weighted Cost |
|---|---|---|---|---|
| Debt | 7.0% | 20,000 | | |
| Preferred Stock | 11.0% | 20,000 | | |
| Common Stock | 14.0% | 60,000 | | |
| **Total** | | **100,000** | **WACC** | |

**Question 1(c):** Calculate WACC from the following data. *(Solved)*

| Capital Component | Cost % | Total Financing | Proportion | Weighted Cost |
|---|---|---|---|---|
| Debt | 7% | 30,000 | 23.07% | 1.61 |
| Preferred Stock | 12% | 40,000 | 30.77% | 3.60 |
| Common Stock | 15% | 60,000 | 46.15% | 6.92 |
| **Total** | | **130,000** | **WACC** | **12.13%** |

**Question 1(d):** Calculate WACC from the following data. *(Student Practice)*

| Capital Component | Cost % | Total Financing |
|---|---|---|
| Debt | 17% | 250,000 |
| Preferred Stock | 12% | 270,000 |
| Common Stock | 19% | 160,000 |
| Retained Earnings | 20% | 150,000 |
| **Total** | | **830,000** |

## Section B: WACC with Project Selection (Midwest Electric — Variations)

**Decision Rule:** Accept a project only if its rate of return **exceeds** the WACC.

**Question 2(a):** Midwest Electric uses only debt and common equity. Interest rate: 10%; capital structure: 45% debt, 55% equity; last dividend: $2; growth rate: 4%; stock price: $20; tax rate: 40%. Projects available: Project A (13%) and Project B (10%).

*Required:*

1. Cost of common equity.
2. Cost of debt (before tax).
3. After-tax cost of debt.
4. WACC.
5. Which projects should Midwest accept?

**Question 2(b):** Same as above but interest rate is 15%; capital structure: 40% debt, 60% equity; tax rate: 40%.

*Required:*

1. Cost of common equity.
2. After-tax cost of debt.
3. WACC.
4. Which projects should Midwest accept?

**Question 3:** Same framework but interest rate is 10%; capital structure: 35% debt, 65% equity; stock price: $25; tax rate: 30%.

*Required:*

1. Cost of common equity.
2. After-tax cost of debt.
3. WACC.
4. Which projects should Midwest accept?

**Question 4:** Same framework but interest rate is 30%; capital structure: 45% debt, 55% equity; last dividend: $5; stock price: $20; tax rate: 40%.

*Required:*

1. Cost of common equity.
2. After-tax cost of debt.
3. WACC.
4. Which projects should Midwest accept?

**Question 5:** Midwest Electric uses debt, common equity, retained earnings, and preferred stock. Interest rate: 10%; capital structure: 30% debt, 50% equity, 10% retained earnings, 10% preferred stock; last dividend (common): $5; growth rate: 4%; stock price: $20; tax rate: 30%; last preferred dividend: $10; preferred stock price: $300. Cost of retained earnings equals cost of equity. Projects available: Project A (13%) and Project B (10%).

*Required:*

1. Cost of common equity.
2. After-tax cost of debt.
3. Cost of preferred stock.
4. WACC.
5. Which projects should Midwest accept?

**Question 6:** Same framework; interest rate: 12%; capital structure: 45% debt, 55% equity; last dividend: $2; growth rate: 4%; stock price: $20; tax rate: 40%. Projects available: Project A (17%) and Project B (15%).

*Required:*

1. Cost of common equity.
2. After-tax cost of debt.
3. WACC.
4. Which projects should Midwest accept?

## Section C: WACC — Solving for an Unknown Component

**Question 7(a):** Percy Motors has a target capital structure of 40% debt and 60% common equity (no preferred stock). YTM on outstanding 20% bonds is 9%; tax rate is 40%; estimated WACC is 9.96%.

*Required:*

1. Cost of debt.
2. After-tax cost of debt.
3. Cost of common equity.

**Question 7(b):** Percy Motors; target capital structure: 40% debt, 60% equity; YTM on outstanding 15% bonds is 10%; tax rate: 30%; estimated WACC is 15%.

*Required:*

1. Cost of debt.
2. After-tax cost of debt.
3. Cost of common equity.

**Question 7(c):** Percy Motors; target capital structure: 40% debt, 60% equity; YTM on bonds: 10%; tax rate: 30%; WACC: 15%; cost of equity: 18%.

*Required:*

1. After-tax cost of debt.

## Section D: Comprehensive WACC Problems

**Question 8:** Patton Paints Corporation has a target capital structure of 40% debt and 60% common equity (no preferred stock). Before-tax cost of debt: 12%; marginal tax rate: 40%; current stock price $P_0$ = \$22.50; last dividend $D_0$ = \$2.00; expected constant growth rate: 7%.

*Required:*

1. Cost of debt.
2. After-tax cost of debt.
3. Cost of common equity.
4. WACC.

**Question 9:** Hook Industries' capital structure consists solely of debt and common equity. Debt can be issued at $r\_d$ = 11%; common stock currently pays $D_0$ = \$2.00; stock price = \$24.75; expected constant dividend growth: 7%; tax rate: 35%; WACC: 13.95%.

*Required:* What percentage of the company's capital structure consists of debt?

**Question 10:** Why should the cost of capital be calculated as a weighted average of the various types of funds a firm generally uses, rather than as the cost of the specific type of capital used in a given year?

**Question 11:** Klose Outfitters Inc. has an optimal capital structure of 60% common equity and 40% debt; tax rate: 40%. To fund an upcoming expansion, the firm has:

- \$2 million of new retained earnings at $r_s = 12\%$
- New common stock up to \$6 million at $r_e = 15\%$
- Up to \$3 million of debt at $r_d = 10\%$
- Additional \$4 million of debt at $r_d = 12\%$

The proposed expansion requires \$5.9 million.

*Required:* What is the WACC for the last dollar raised to complete the expansion?

# Capital Budgeting Techniques

## Basic Concepts

| Term | Definition |
|---|---|
| **Discounted Cash Flow (DCF)** | Any method of investment project evaluation that adjusts cash flows over time for the time value of money. |
| **Payback Period (PBP)** | The period of time required for the cumulative expected cash flows from an investment project to equal the initial cash outflow. |
| **Discounted Payback Period** | The time required for cumulative *discounted* cash flows to recover the initial investment. |
| **Net Present Value (NPV)** | The present value of a project's net cash flows minus the initial cash outflow. |
| **Internal Rate of Return (IRR)** | The discount rate that equates the present value of future net cash flows with the project's initial cash outflow. |
| **Hurdle Rate** | The minimum required rate of return on an investment; the rate at which a project becomes acceptable. |
| **Profitability Index (PI)** | The ratio of the present value of future net cash inflows to the project's initial cash outflow. |
| **NPV Profile** | A graph showing the relationship between a project's NPV and the discount rate employed. |
| **Independent Project** | A project whose acceptance or rejection does not affect the acceptance of other projects. |
| **Dependent Project** | A project whose acceptance depends on the acceptance of one or more other projects. |
| **Mutually Exclusive Project** | A project whose acceptance precludes the acceptance of one or more alternative projects. |

## Key Formulae

**NPV** = PV of Cash Inflows − PV of Cash Outflows

**Profitability Index (PI)** = PV of Cash Inflows / PV of Cash Outflows

**Decision Rules:**

- Accept if NPV > 0; Reject if NPV < 0
- Accept if PI > 1; Reject if PI < 1

- Accept if IRR > Hurdle Rate

# Payback Period — Activities

### Activity I *(Solved)*

Initial investment: Rs. 50,000

| Year | Cash Flows | Remaining Balance | Payback Year |
|---|---|---|---|
| 0 | (50,000) | (50,000) | 0 |
| 1 | 20,000 | (30,000) | 1 |
| 2 | 20,000 | (10,000) | 1 |
| 3 | 10,000 | 0 | 1 |
| 4 | 10,000 | — | |
| 5 | 10,000 | — | |
| | | **Payback Period** | **3 Years** |

### Activity II *(Student Practice)*

Initial investment: Rs. 70,000

| Year | Cash Flows | Remaining Balance | Payback Year |
|---|---|---|---|
| 0 | (70,000) | (70,000) | 0 |
| 1 | 20,000 | | |
| 2 | 20,000 | | |
| 3 | 10,000 | | |
| 4 | 10,000 | | |
| 5 | 10,000 | | |
| | | **Payback Period** | |

### Activity III *(Solved)*

Initial investment: Rs. 45,000

| Year | Cash Flows | Remaining Balance | Payback Year |
|---|---|---|---|
| 0 | (45,000) | (45,000) | 0 |
| 1 | 20,000 | 25,000 | 1 |
| 2 | 20,000 | 5,000 | 1 |
| 3 | 20,000 | 5,000/20,000 = 0.25 | 0.25 |
| 4 | 10,000 | — | |
| 5 | 10,000 | — | |
| | | **Payback Period** | **2.25 Years** |

### Activity IV *(Student Practice)*

Initial investment: Rs. 145,000

| Year | Cash Flows | Remaining Balance | Payback Year |
|---|---|---|---|
| 0 | (145,000) | (145,000) | 0 |
| 1 | 25,000 | | |
| 2 | 45,000 | | |
| 3 | 20,000 | | |
| 4 | 90,000 | | |
| 5 | 10,000 | | |
| | | **Payback Period** | |

### Payback Period — Advantages, Disadvantages & Assumptions

**Advantages:** Easy to calculate and communicate; clearly conveys the time required to recover an investment.

**Disadvantages:** Does not consider cash flows occurring after the payback period; ignores the time value of money.

**Assumption:** Cash flows within a year are assumed to occur evenly throughout the year.

# Present Value Method — Activities

### Q5 — NPV @ 10% *(Solved)*

| Year | Cash Flows | PVF @ 10% | Present Value |
|---|---|---|---|
| 0 | (70,000) | 1.000 | (70,000) |
| 1 | 20,000 | 0.909 | 18,180 |
| 2 | 20,000 | 0.826 | 16,520 |
| 3 | 20,000 | 0.751 | 15,020 |
| 4 | 20,000 | 0.683 | 13,660 |
| 5 | 10,000 | 0.620 | 6,200 |
| | | **NPV** | **(420)** |

**Decision:** NPV is **negative** → **Reject** the project.

**PI** = 69,580 / 70,000 = **0.994** → PI < 1 → **Reject**

## Q5 — NPV @ 5% *(Solved)*

| Year | Cash Flows | PVF @ 5% | Present Value |
|---|---|---|---|
| 0 | (70,000) | 1.0000 | (70,000) |
| 1 | 20,000 | 0.9524 | 19,048 |
| 2 | 20,000 | 0.9070 | 18,140 |
| 3 | 20,000 | 0.8638 | 17,276 |
| 4 | 20,000 | 0.8227 | 16,454 |
| 5 | 10,000 | 0.7835 | 7,835 |
| | | **NPV** | **8,753** |

**Decision:** NPV is **positive** → **Accept** the project.

## Q5 — NPV @ 7% *(Solved)*

| Year | Cash Flows | PVF @ 7% | Present Value |
|---|---|---|---|
| 0 | (70,000) | 1.000 | (70,000) |
| 1 | 20,000 | 0.934 | 18,680 |
| 2 | 20,000 | 0.872 | 17,440 |
| 3 | 20,000 | 0.815 | 16,300 |
| 4 | 20,000 | 0.761 | 15,220 |
| 5 | 10,000 | 0.711 | 7,110 |

| Year | Cash Flows | PVF @ 7% | Present Value |
|---|---|---|---|
| | | **NPV** | **4,750** |

**Decision:** NPV is **positive** → **Accept** the project.

**PI** = 74,750 / 70,000 = **1.068** → PI > 1 → **Accept**

### Q6 — NPV @ 15% *(Solved)*

| Year | Cash Flows | PVF @ 15% | Present Value |
|---|---|---|---|
| 0 | (70,000) | 1.000 | (70,000) |
| 1 | 20,000 | 0.869 | 17,380 |
| 2 | 20,000 | 0.755 | 15,100 |
| 3 | 20,000 | 0.656 | 13,120 |
| 4 | 20,000 | 0.570 | 11,400 |
| 5 | 10,000 | 0.495 | 4,950 |
| | | **NPV** | **(8,050)** |

**Decision:** NPV is **negative** → **Reject** the project.

**PI** = 61,950 / 70,000 = **0.885** → PI < 1 → **Reject**

### Q7 — NPV @ 9% *(Student Practice)*

| Year | Cash Flows | PVF @ 9% | Present Value |
|---|---|---|---|
| 0 | (70,000) | | |
| 1 | 20,000 | | |
| 2 | 20,000 | | |
| 3 | 20,000 | | |
| 4 | 20,000 | | |
| 5 | 10,000 | | |
| | | **NPV** | |

# Decision Criteria Summary

| Method | Accept | Reject |
|---|---|---|
| **NPV** | NPV > 0 | NPV < 0 |
| **Profitability Index** | PI > 1 | PI < 1 |
| **IRR** | IRR > Hurdle Rate | IRR < Hurdle Rate |
| **Payback Period** | PBP < Target Period | PBP > Target Period |

# Conceptual Questions

**Q1.** Briefly explain the project evaluation techniques and methods.

**Q2.** Briefly explain the advantages and disadvantages of the payback period.

**Q3.** What is meant by the present value method of project evaluation?

**Q4.** Briefly explain the advantages and disadvantages of the present value method of project evaluation.

**Q5.** Briefly explain the profitability index method of project evaluation.

# Practice Problems

**Question 1:** Carbide Chemical Company is considering replacing two old machines with a new, more efficient machine. The relevant after-tax incremental operating cash flows are as follows.

**Payback Period** *(Solved)*

| Year | Cash Flow ($) | Remaining Balance | Payback Year |
|---|---|---|---|
| 0 | (404,424) | 404,424 | 0 |
| 1 | 86,890 | 317,534 | 1 |
| 2 | 106,474 | 211,060 | 1 |
| 3 | 91,612 | 119,448 | 1 |
| 4 | 84,801 | 34,647 | 1 |
| 5 | 84,801 | 34,647/84,801 | 0.41 |
| 6 | 75,400 | — | |
| 7 | 66,000 | — | |

| Year | Cash Flow ($) | Remaining Balance | Payback Year |
|---|---|---|---|
| 8 | 92,400 | **Payback Period** | **4.41 Years** |

**NPV @ 14%** *(Solved)*

| Year | Cash Flow ($) | PVF @ 14% | Present Value ($) |
|---|---|---|---|
| 0 | (404,424) | 1.0000 | (404,424) |
| 1 | 86,890 | 0.8772 | 76,220 |
| 2 | 106,474 | 0.7695 | 81,932 |
| 3 | 91,612 | 0.6750 | 61,838 |
| 4 | 84,801 | 0.5925 | 50,211 |
| 5 | 84,801 | 0.5194 | 44,046 |
| 6 | 75,400 | 0.4556 | 34,352 |
| 7 | 66,000 | 0.3996 | 26,374 |
| 8 | 92,400 | 0.3506 | 32,395 |
| | | **NPV** | **+2,943** |

**Decision:** NPV is **positive** → **Accept** the project.

**Discounted Payback Period** *(Solved)*

| Year | Cash Flow ($) | Discounted CF | Remaining Balance | Payback Year |
|---|---|---|---|---|
| 0 | (404,424) | — | 404,424 | 0 |
| 1 | 86,890 | 76,220 | 328,204 | 1 |
| 2 | 106,474 | 81,932 | 246,272 | 1 |
| 3 | 91,612 | 61,838 | 184,434 | 1 |
| 4 | 84,801 | 50,211 | 134,223 | 1 |
| 5 | 84,801 | 44,046 | 90,177 | 1 |
| 6 | 75,400 | 34,352 | 55,826 | 1 |
| 7 | 66,000 | 26,374 | 29,452 | 1 |
| 8 | 92,400 | 32,395 | 29,452/32,395 | 0.909 |
| | | | **Discounted Payback Period** | **7.909 Years** |

*Required:*

1. What is the project's payback period?
2. What is the project's discounted payback period?
3. What is the project's NPV at 14%? Is the project acceptable?
4. What is the profitability index? Is the project acceptable?

**Question 2:** Briarcliff Stove Company is considering a new product line requiring a cash investment of $700,000 at time 0. After-tax cash inflows: $250,000 (Year 1), $300,000 (Year 2), $350,000 (Year 3), and $400,000 per year from Year 4 through Year 10.

**Payback Period** *(Solved)*

| Year | Cash Flow ($) | Remaining Balance | Payback Year |
|---|---|---|---|
| 0 | (700,000) | 700,000 | 0 |
| 1 | 250,000 | 450,000 | 1 |
| 2 | 300,000 | 150,000 | 1 |
| 3 | 350,000 | 150,000/350,000 | 0.43 |
| 4–10 | 400,000 | — | |
| | | **Payback Period** | **2.43 Years** |

**NPV @ 15%** *(Solved)*

| Year | Cash Flow ($) | PVF @ 15% | Present Value ($) |
|---|---|---|---|
| 0 | (700,000) | 1.0000 | (700,000) |
| 1 | 250,000 | 0.8696 | 217,400 |
| 2 | 300,000 | 0.7561 | 226,830 |
| 3 | 350,000 | 0.6575 | 230,125 |
| 4 | 400,000 | 0.5718 | 228,720 |
| 5 | 400,000 | 0.4972 | 198,880 |
| 6 | 400,000 | 0.4323 | 172,920 |
| 7 | 400,000 | 0.3759 | 150,360 |
| 8 | 400,000 | 0.3269 | 130,760 |
| 9 | 400,000 | 0.2843 | 113,720 |
| 10 | 400,000 | 0.2472 | 98,880 |
| | | **NPV** | **+1,068,595** |

**Decision:** NPV is **positive** → **Accept** the project.

**PI** = 1,768,595 / 700,000 = **2.52** → Accept

**IRR Calculation** *(Solved — Interpolation between 15% and 50%)*

| Year | CF ($) | PVF @ 15% | PV @ 15% | PVF @ 50% | PV @ 50% |
|---|---|---|---|---|---|
| 0 | (700,000) | 1.0000 | (700,000) | 1.0000 | (700,000) |

| Year | CF ($) | PVF @ 15% | PV @ 15% | PVF @ 50% | PV @ 50% |
|---|---|---|---|---|---|
| 1 | 250,000 | 0.8696 | 217,400 | 0.6667 | 166,675 |
| 2 | 300,000 | 0.7561 | 226,830 | 0.4445 | 133,350 |
| 3 | 350,000 | 0.6575 | 230,125 | 0.2963 | 103,705 |
| 4 | 400,000 | 0.5718 | 228,720 | 0.1976 | 79,040 |
| 5 | 400,000 | 0.4972 | 198,880 | 0.1317 | 52,680 |
| 6 | 400,000 | 0.4323 | 172,920 | 0.0878 | 35,120 |
| 7 | 400,000 | 0.3759 | 150,360 | 0.0585 | 23,400 |
| 8 | 400,000 | 0.3269 | 130,760 | 0.0390 | 15,600 |
| 9 | 400,000 | 0.2843 | 113,720 | 0.0260 | 10,400 |
| 10 | 400,000 | 0.2472 | 98,880 | 0.0173 | 6,920 |
| | | **NPV** | **+1,068,595** | **NPV** | **(73,110)** |

**IRR ≈ 47.24%**

*Required:*

1. What is the project's payback period?
2. What is the NPV at 15%? Is the project acceptable?
3. What would be the case if the required rate of return were 10%?
4. What is the project's IRR?
5. Calculate the PI and give your recommendation at 15%.
6. Calculate the PI and give your recommendation at 10%.

**Question 3:** Briarcliff Stove Company — same as Question 2, but with an additional cash investment of $1,000,000 in Year 1 (in addition to $700,000 at time 0). After-tax cash inflows begin in Year 2.

*Required:*

1. What is the project's payback period?
2. What is the NPV at 15%? Is the project acceptable?
3. What would be the case if the required rate of return were 10%?
4. What is the project's IRR?
5. Calculate the PI at 15% and give your recommendation.

**Question 4:** Carbide Chemical Company — same replacement scenario as Question 1, but with simplified uniform cash flows of $100,000 per year for 5 years against an initial outlay of $404,424.

**Payback Period** *(Student Practice)*

| Year | Cash Flow ($) | Remaining Balance | Payback Year |
|---|---|---|---|
| 0 | (404,424) | 404,424 | 0 |
| 1 | 100,000 | | 1 |
| 2 | 100,000 | | |
| 3 | 100,000 | | |
| 4 | 100,000 | | |
| 5 | 100,000 | | |

**NPV @ 14%** *(Student Practice)*

| Year | Cash Flow ($) | PVF @ 14% | Present Value ($) |
|---|---|---|---|
| 0 | (404,424) | | |
| 1 | 100,000 | | |
| 2 | 100,000 | | |
| 3 | 100,000 | | |
| 4 | 100,000 | | |
| 5 | 100,000 | | |

*Required:*

1. What is the project's payback period?
2. What is the project's discounted payback period?
3. What is the NPV at 14%? Is the project acceptable?
4. What is the profitability index? Is the project acceptable?

**Question 5:** Lobers, Inc. has two investment proposals with the following characteristics.

| Period | Project A — CF ($) | Period | Project B — CF ($) |
|---|---|---|---|
| 0 | (9,000) | 0 | (12,000) |
| 1 | 5,000 | 1 | 5,000 |
| 2 | 4,000 | 2 | 5,000 |
| 3 | 3,000 | 3 | 8,000 |

*Required:* For each project, compute:

1. Payback period.

2. Discounted payback period at 15%.
3. NPV at 15%.
4. Profitability index at 15%.
5. Internal rate of return (IRR).

**Question 6:** Zaire Electronics can make one of two investments at time 0.

| Project | Investment | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Year 6 | Year 7 |
|---|---|---|---|---|---|---|---|---|
| A | $28,000 | 8,000 | 8,000 | 8,000 | 8,000 | 8,000 | 8,000 | 8,000 |
| B | $20,000 | 5,000 | 5,000 | 6,000 | 6,000 | 7,000 | 7,000 | 7,000 |

*Required:* Assuming a required rate of return of 14%, determine for each project:

1. Payback period.
2. NPV.
3. Profitability index.

**Question 7:** Project ZZ cash flows:

| Year | Cash Flow ($) |
|---|---|
| 0 | (100,000) |
| 1–4 | 35,027 each |

*Required:*

1. Payback period.
2. NPV at 10%.
3. NPV at 16%.
4. PI at 10%.
5. PI at 16%.
6. IRR.

**Question 8:** Project XX cash flows:

| Year | Cash Flow ($) |
|---|---|
| 0 | (200,000) |
| 1–4 | 65,000 each |

*Required:*

1. Payback period.
2. NPV at 10%.
3. NPV at 15%.
4. PI at 10%.
5. PI at 15%.
6. IRR.

**Question 9:** Consider these two projects:

| Project | $C_0$ | $C_1$ | $C_2$ | $C_3$ |
|---|---|---|---|---|
| A | −$18 | +$10 | +$10 | +$10 |
| B | −$50 | +$25 | +$25 | +$25 |

*Required:*

1. Which project has the higher NPV at a 10% discount rate?
2. Which has the higher profitability index?
3. Which is most attractive to a firm that can raise unlimited funds?
4. Which is most attractive to a firm with limited capital?

**Question 10:** Consider the following projects:

| Project | $C_0$ | $C_1$ | $C_2$ |
|---|---|---|---|
| A | −$2,100 | +$2,000 | +$1,200 |
| B | −$2,100 | +$1,440 | +$1,728 |

*Required:*

1. Calculate the PI for A and B assuming a 20% opportunity cost of capital.
2. Use the PI rule to determine which project(s) to accept: (i) if both can be undertaken, and (ii) if only one can be undertaken.

**Question 11:**

| Project | $C_0$ | $C_1$ | $C_2$ | $C_3$ | $C_4$ |
|---|---|---|---|---|---|
| A | −5,000 | +1,000 | +1,000 | +3,000 | 0 |

| Project | $C_0$ | $C_1$ | $C_2$ | $C_3$ | $C_4$ |
|---|---|---|---|---|---|
| B | −1,000 | 0 | +1,000 | +2,000 | +3,000 |
| C | −5,000 | +1,000 | +1,000 | +3,000 | +5,000 |

*Required:*

1. What is the payback period for each project?
2. If the opportunity cost of capital is 10%, which projects have positive NPVs?

**Question 12:** A new computer system requires an initial outlay of $20,000 and will increase cash flows by $4,000 per year for 8 years.

*Required:*

1. Should the project be accepted if the required return is 12%?
2. Is the system worth installing if the required return is 9%?
3. What if the required return is 14%? How high can the discount rate be before the project is rejected?

**Question 13:** Project L costs $65,000; expected cash inflows are $12,000 per year for 9 years; WACC is 9%.

*Required:*

1. What is the project's NPV?
2. What is the project's IRR?
3. What is the project's payback period?

**Question 14:** Your division is considering two projects (in millions):

| Period | Project A | Project B |
|---|---|---|
| 0 | −25 | −20 |
| 1 | 5 | 10 |
| 2 | 10 | 9 |
| 3 | 17 | 6 |

*Required:*

1. What are the NPVs of each project at WACCs of 5%, 10%, and 15%?
2. What are the IRRs of each project?

**Question 15:** Consider projects A and B:

| Project | $C_0$ | $C_1$ | $C_2$ | NPV @ 10% |
|---|---|---|---|---|
| A | −30,000 | 21,000 | 21,000 | +$6,446 |
| B | −50,000 | 33,000 | 33,000 | +$7,273 |

*Required:*

1. Calculate the IRR for A and B.
2. Which project does the IRR rule suggest is best?
3. Which project is actually the better choice?

**Question 16:** A project costs $3,000 to install and provides annual cash flows of $800 for 6 years. Discount rate is 10%.

*Required:*

1. Is this project worth pursuing?
2. How high can the discount rate be before the project is rejected?

**Question 17:** What is the profitability index of a project costing $10,000 that provides cash flows of $3,000 in Years 1 and 2, and $5,000 in Years 3 and 4? The discount rate is 10%.

*Required:* What is the profitability index?

# Budgeting — Comprehensive Practice Notes

## Part I: Production & Purchases Budgets

### Key Formula

**Production / Purchases Budget:** Closing Stock Required **+ Stock Sold / Used** = Total Stock Required **− Opening Stock Available** = **Total Stock to be Produced / Purchased**

### Exercise 1 — Purchases Budget *(Solved)*

Noardstone's Department Store — April sales: $75,000; opening inventory: $50,000; closing inventory: $42,500; cost of goods sold: 70% of sales.

| | Amount ($) |
|---|---:|
| Closing Stock Required | 42,500 |
| Add: Cost of Goods Sold (75,000 × 70%) | 52,500 |
| **Total Stock Required** | **95,000** |
| Less: Opening Stock Available | (50,000) |
| **Total Purchases Required** | **45,000** |

### Exercise 2 — Production Budget: OGDCL *(Partially Solved)*

Closing stock policy: 10% of next year's expected sales.

**Year 2014** *(Solved)*

| | Litres |
|---|---:|
| Closing Stock Required (34,000 × 10%) | 3,400 |
| Add: Stock Sold | 26,000 |
| **Total Stock Required** | **29,400** |
| Less: Opening Stock | (2,850) |
| **Total Production** | **26,550** |

**Year 2015** *(Student Practice)*

| | **Litres** |
|---|---|
| Closing Stock Required (28,000 × 10%) | |
| Add: Stock Sold | |
| **Total Stock Required** | |
| Less: Opening Stock | |
| **Total Production** | |

**Year 2016** *(Student Practice)*

| | **Litres** |
|---|---|
| Closing Stock Required (22,000 × 10%) | |
| Add: Stock Sold | |
| **Total Stock Required** | |
| Less: Opening Stock | |
| **Total Production** | |

## Exercise 3 — Production Schedule: Hike n' Run Company *(Partially Solved)*

Uniform production: 72,000 pairs/quarter. Opening inventory (Dec 31, 2009): 38,400 pairs.

**Quarter 1** *(Solved)*

| | **Pairs** |
|---|---|
| Closing Stock Required | 0 |
| Add: Stock Sold | 76,800 |
| **Total Required** | **76,800** |
| Less: Opening Stock | (38,400) |
| **Total Production** | **38,400** |

**Quarter 2** *(Solved)*

| | **Pairs** |
|---|---|
| Closing Stock Required | 0 |
| Add: Stock Sold | 62,400 |
| **Total Required** | **62,400** |
| Less: Opening Stock | (0) |

| | Pairs |
|---|---|
| **Total Production** | **62,400** |

**Quarter 3** *(Student Practice)*

| | Pairs |
|---|---|
| Closing Stock Required | |
| Add: Stock Sold | |
| **Total Required** | |
| Less: Opening Stock | |
| **Total Production** | |

**Quarter 4** *(Student Practice)*

| | Pairs |
|---|---|
| Closing Stock Required | |
| Add: Stock Sold | |
| **Total Required** | |
| Less: Opening Stock | |
| **Total Production** | |

## Exercise 4 — Production & Labour Budget: Landy

Landy manufactures Products A and B using Grade Q labour. Current inventory: 800 units of A, 1,200 units of B (250 units of B scrapped). Budgeted sales: A = 3,000 units, B = 4,000 units. Closing inventory policy: 3 months' sales. Original labour hours: 2 hrs/unit A, 3 hrs/unit B at $2.50/hr. Revised: wage rate increased by $0.50/hr; production times reduced by 20%.

*Required:* Prepare the production budget and direct labour budget for 20X3.

## Exercise 5 — Production & Raw Materials Budget: W Wallace Ltd

Product Y requires: Material A (4 kg @ £3.75), Material B (3 kg @ £4.80), Material C (5 kg @ £3.10), Direct Labour (9 hrs @ £9.00/hr). Sales estimates:

| Month | Units |
|---|---|
| January | 500 |
| February | 450 |

| Month | Units |
|---|---|
| March | 550 |
| April | 540 |

Closing stock policy: 50% of next month's sales. Quality control rejection rate: 10% of production written off. Raw material stocks at Jan 1: A = 800 kg, B = 500 kg, C = 700 kg. Stocks to increase by 10% in January and each month thereafter. 25% of total attendance time is on production support; staff paid on attendance time basis. Selling price: £275/unit.

*Required:* For the quarter January–March 2010, prepare:

1. Sales budget (quantity and value).
2. Production budget (units).
3. Raw material usage budget (kg).
4. Raw material purchases budget (kg and value).

### Exercise 6 — Production & Labour Budget: Truro

Sales: 700 units; opening finished goods: 50 units; closing finished goods: 70 units. 10% of finished work scrapped. Standard direct labour: 3 hrs/unit. Labour productivity ratio: 80%. 18 direct operatives averaging 144 working hours each in Period 3.

*Required:*

1. Production budget.
2. Direct labour budget.
3. Comment on any problem revealed and suggest how it might be overcome.

# Part II: Budgeted Income Statement

### Key Formula

**Sales** − Cost of Goods Sold / Production Costs = **Gross Profit** − Operating Expenses (Selling, Admin, etc.) = **Net Income (before tax)**

### Activity 1 — Budgeted Income Statement *(Solved)*

**OGDCL — Income Statement** *For the Month of February 2015*

| | **Rs.** |
|---|---|
| Sales | 10,000 |
| Less: Production Costs | (7,000) |
| **Gross Profit** | **3,000** |
| Less: Other Expenses | (1,000) |
| **Net Income** | **2,000** |

## Activity 2 — Budgeted Income Statement *(Student Practice)*

**OGDCL — Income Statement** *For the Month of February 2012*

| | **Rs.** |
|---|---|
| Sales | 10,000 |
| Less: Production Costs | |
| **Gross Profit** | |
| Less: Other Expenses | |
| Marketing Expenses | |
| Distribution Expenses | |
| **Net Income** | |

## Activity 3 — Budgeted Income Statement *(Student Practice)*

**OGDCL — Income Statement** *For the Month of February 2012*

| | **Rs.** |
|---|---|
| Sales | |
| Other Income | |
| **Total Income** | |
| **Less: Production Costs** | |
| — Materials | |
| — Labour | |
| — Use of Facilities | |
| **Gross Profit** | |
| **Less: Operating Expenses** | |
| — Transportation | |

| | Rs. |
|---|---|
| — Administration | |
| — Tax | |
| — Other Costs of Selling & Delivery | |
| **Net Income** | |

### Exercise 7 — Budgeted Income: DePaul Company *(Solved)*

Sales: 25,000 units @ $6.00; production cost: $1.80/unit; variable S&A: $0.60/unit; fixed S&A: $60,000.

*Required:* Compute budgeted income before tax.

### Exercise 8 — Budgeted Income: Skaters Plus *(Student Practice)*

Sales: 90,000 skateboards @ $36; production cost: $14.40/unit; variable S&A: $7.20/unit; fixed S&A: $604,800/quarter.

*Required:* What are the budgeted earnings for next quarter?

# Part III: Cash Budget

### Basic Concept

A **cash budget** is a table showing cash receipts, disbursements, and balances over a given period. It enables managers to anticipate cash surpluses and deficits in advance.

**General Structure:**

| Section | Items |
|---|---|
| **Cash Receipts** | All inflows (sales collections, salary, dividends, etc.) |
| **Cash Payments** | All outflows (purchases, wages, expenses, taxes, etc.) |
| **Excess / (Deficit)** | Total Receipts − Total Payments |
| **Opening Balance** | Closing balance from the prior period |
| **Closing Balance** | Opening Balance + Excess / (Deficit) |

**Note:** Depreciation is a **non-cash** item and must **never** be included in a cash budget.

### Activity 1 — Basic Cash Budget *(Solved)*

| | **Jan** | **Feb** | **Mar** | **Apr** |
|---|---|---|---|---|
| **Cash Receipts** | | | | |
| Pay Received | 100 | 100 | 100 | 100 |
| **Total Cash Received** | **100** | **100** | **100** | **100** |
| **Cash Payments** | | | | |
| Kitchen Expenses | 5 | 10 | 10 | 15 |
| **Total Payments** | **5** | **10** | **10** | **15** |
| **Excess Cash Balance** | **95** | **90** | **90** | **85** |
| Opening Balance | 10 | 105 | 195 | 285 |
| Excess / (Deficit) | 95 | 90 | 90 | 85 |
| **Closing Balance** | **105** | **195** | **285** | **370** |

### Activity 2 — Extended Cash Budget *(Solved)*

| | **Jan** | **Feb** | **Mar** | **Apr** |
|---|---|---|---|---|
| **Cash Receipts** | | | | |
| Pay Received | 100 | 100 | 100 | 100 |
| **Total Cash Received** | **100** | **100** | **100** | **100** |
| **Cash Payments** | | | | |
| Electricity Bills | 5 | 10 | 5 | 5 |
| Sui-gas Bills | 5 | 5 | 5 | 10 |
| Kitchen Expenses | 2 | 5 | 5 | 5 |
| **Total Payments** | **12** | **20** | **15** | **20** |
| **Excess / (Deficit)** | **88** | **80** | **85** | **80** |
| Opening Balance | 10 | 98 | 178 | 263 |
| Excess / (Deficit) | 88 | 80 | 85 | 80 |
| **Closing Balance** | **98** | **178** | **263** | **343** |

### Activity 3 — Deficit Cash Budget *(Solved)*

| | **Jan** | **Feb** | **Mar** | **Apr** |
|---|---|---|---|---|
| **Cash Receipts** | | | | |
| Pay Received | 100 | 100 | 100 | 100 |
| **Total Cash Received** | **100** | **100** | **100** | **100** |
| **Cash Payments** | | | | |
| Electricity Bills | 15 | 10 | 50 | 50 |
| Sui-gas Bills | 15 | 50 | 50 | 10 |
| Kitchen Expenses | 12 | 50 | 50 | 15 |
| Purchase of Bicycle | — | 400 | — | — |
| **Total Payments** | **42** | **510** | **150** | **75** |
| **Excess / (Deficit)** | **58** | **(410)** | **(50)** | **25** |
| Opening Balance | 10 | 68 | (342) | (392) |
| Excess / (Deficit) | 58 | (410) | (50) | 25 |
| **Closing Balance** | **68** | **(342)** | **(392)** | **(367)** |

### Activity 4 — Cash Budget: Receipts > Payments *(Student Practice — Hypothetical Values)*

| | **Jan** | **Feb** | **Mar** | **Apr** |
|---|---|---|---|---|
| **Cash Receipts** | | | | |
| Funds Received | | | | |
| **Total Cash Received** | | | | |
| **Cash Payments** | | | | |
| Wages | | | | |
| Electricity Bills | | | | |
| Sui-gas Bills | | | | |
| Transportation | | | | |
| Purchase of Car | | | | |
| **Total Payments** | | | | |
| **Excess / (Deficit)** | | | | |
| Opening Balance | | | | |
| Excess / (Deficit) | | | | |
| **Closing Balance** | | | | |

### Activity 5 — Cash Budget for Mr. Dawood Sulemen *(Student Practice — Hypothetical Values)*

| | Jan | Feb | Mar | Apr |
|---|---|---|---|---|
| **Cash Receipts** | | | | |
| Salary Received | | | | |
| Dividend Income | | | | |
| Rent from Property | | | | |
| Bonus *(April only)* | — | — | — | |
| Proceeds from Sale of Plot *(Feb only)* | — | | — | — |
| **Total Cash Received** | | | | |
| **Cash Payments** | | | | |
| Wages of Servants | | | | |
| Grocery / Kitchen Expenses | | | | |
| Children's School Fee | | | | |
| House Maintenance *(Jan only)* | | — | — | — |
| Electricity Bills | | | | |
| Sui-gas Bills | | | | |
| Transportation | | | | |
| Purchase of Car *(March only)* | — | — | | — |
| **Total Payments** | | | | |
| **Excess / (Deficit)** | | | | |
| Opening Balance | | | | |
| Excess / (Deficit) | | | | |
| **Closing Balance** | | | | |

## Cash Budget — Practice Problems

**Question 1:** Prepare a cash budget for April, May, and June.

| Month | Sales | Purchases | Wages | Expenses |
|---|---|---|---|---|
| January *(Actual)* | 80,000 | 45,000 | 20,000 | 5,000 |
| February *(Actual)* | 80,000 | 40,000 | 18,000 | 6,000 |
| March *(Actual)* | 75,000 | 42,000 | 22,000 | 6,000 |
| April *(Budget)* | 90,000 | 50,000 | 24,000 | 6,000 |
| May *(Budget)* | 85,000 | 45,000 | 20,000 | 6,000 |
| June *(Budget)* | 80,000 | 35,000 | 18,000 | 5,000 |

*Additional information:* 10% of purchases and 20% of sales are cash; credit sales collected after one month; credit purchases paid after one month; wages paid half-monthly; rent Rs. 500/month (excluded from expenses); expenses paid in month incurred; opening balance April 1: Rs. 15,000.

**Question 2:** Same data as Question 1 with revised payment terms: 20% of sales for cash (remainder collected next month); wages paid the month after earned; all other conditions same. Opening balance April 1: Rs. 15,000.

**Question 3:** Focus Group — prepare a cash budget for March through June.

| Month | Sales (Rs.) | Purchases (Rs.) | Wages (Rs.) |
|---|---|---|---|
| February | 1,80,000 | 1,24,800 | 12,000 |
| March | 1,92,000 | 1,44,000 | 14,000 |
| April | 1,08,000 | 2,43,000 | 11,000 |
| May | 1,74,000 | 2,46,000 | 10,000 |
| June | 1,26,000 | 2,68,000 | 15,000 |

*Additional information:* 50% cash sales; 50% collected next month; creditors paid in month of purchase; wages paid in month incurred; opening balance March 1: Rs. 25,000.

**Question 4:** Same base data as Question 3, plus January (Sales: 1,00,000; Purchases: 1,00,000; Wages: 15,000). Prepare cash budget for March–June. *Additional information:* 50% cash sales; 20% collected next month; 30% collected two months later; 50% of creditors paid in purchase month, 50% next month; 50% of wages paid in month incurred, 50% next month; opening balance: Rs. 1,50,000.

**Question 5:** Same base data as Question 3. Prepare cash budget for March–June. *Additional information:* 50% cash sales; 40% collected next month; 10% uncollectable; creditors paid in purchase month; depreciation March: Rs. 7,000 *(exclude — non-cash)*; wages paid in month incurred; opening balance March 1: Rs. 25,000.

**Question 6:** Same data as Question 4 but wages split differently: 50% paid in month incurred; 30% paid next month; 20% paid two months later. Opening balance: Rs. 20,000. Prepare cash budget for March, April, and May only.

**Question 7:** Galaxy Lighting Company.

| Month | Sales |
|---|---|
| August *(Actual)* | $400,000 |
| September | $350,000 |
| October | $500,000 |
| November | $400,000 |

*Additional information:* All sales on account; 60% collected in month of sale; 40% collected next month; selling price $30/unit; purchase cost $18/unit; 60% of purchases paid in purchase month; 40% next month; end-of-month inventory: 1,000 units + 10% of next month's unit sales; opening balance September 1: $100,000.

*Required:* Cash receipts schedule for September and October; purchases budget for August, September, and October.

**Question 8:** Galaxy Lighting Company — same structure as Question 7 with revised figures: August sales $700,000; September $450,000; October $600,000; November $600,000; purchase cost $20/unit. Additionally, selling and administrative expenses paid in cash are $120,000/month.

*Required:* Cash receipts schedule for September and October; purchases budget for August, September, and October; monthly cash budget for September and October.

**Question 9:** XY Ltd.

| Month | Wages (Rs.'000) | Materials (Rs.'000) | Overhead (Rs.'000) | Sales (Rs.'000) |
|---|---|---|---|---|
| February | 6 | 20 | 10 | 30 |
| March | 8 | 30 | 12 | 40 |
| April | 10 | 25 | 16 | 60 |
| May | 9 | 35 | 14 | 50 |
| June | 12 | 30 | 18 | 70 |
| July | 10 | 25 | 16 | 60 |
| August | 9 | 25 | 14 | 50 |

*Additional information:* Opening balance May 31: Rs. 22,000; wages paid in month incurred; materials paid 3 months after receipt; credit customers pay 2 months after delivery; overhead includes Rs. 2,000/month depreciation *(non-cash)*; remaining overhead paid with 1-month delay; 10% cash sales, 90% credit; 5% commission on credit sales paid the following month *(not in overhead)*; loan repayment Rs. 25,000 due June 30; new machine July: Rs. 45,000 (Rs. 15,000 on delivery; Rs. 15,000 each of the following two months).

*Required:* Cash budget for June, July, and August.

**Question 10:** Mansel Ltd — publishing company. Selling price: $15/book. Variable costs: materials $5, labour $4, overhead $2.

| **Month** | **Nov** | **Dec** | **Jan** | **Feb** | **Mar** | **Apr** | **May** | **Jun** | **Jul** | **Aug** |
|---|---|---|---|---|---|---|---|---|---|---|
| Books | 1,000 | 1,000 | 1,000 | 1,250 | 1,500 | 2,000 | 1,900 | 2,200 | 2,200 | 2,300 |

*Additional information:* Collections: 40% one month after sale, 60% two months after sale; books produced 2 months before sale; material creditors paid 2 months after production; variable overheads paid month after production (increase 25% from April); wages: 75% in production month, 25% next month (12.5% increase from March 1); property sold May: $25,000; press purchased May: $10,000; depreciation: $1,000/month rising to $1,500 post-purchase; corporation tax $10,000 due March; opening balance December 31: $1,500.

*Required:* Cash budget for January 1 to June 30.

**Question 11:** Company budget data — November 2008 to June 2009.

| | **Nov** | **Dec** | **Jan** | **Feb** | **Mar** | **Apr** | **May** | **Jun** |
|---|---|---|---|---|---|---|---|---|
| Sales (000) | 60 | 90 | 100 | 120 | 130 | 140 | — | 170 |
| Purchases (000) | 30 | 50 | 70 | 80 | 100 | 110 | 120 | 140 |
| Wages (000) | 8 | 15 | 18 | 18 | 20 | 22 | 24 | 30 |
| Overheads (000) | 8 | 8 | 12 | 15 | 20 | 20 | 20 | 20 |
| Dividends (000) | — | 18 | — | — | — | — | 36 | — |
| Capital Expenditure (000) | — | — | 40 | — | — | — | 50 | — |

*Additional information:* Sales: 50% cash, 50% credit (collected 1 month later); purchases paid 1 month after purchase; wages: 80% in month incurred, 20% next month; overheads paid month after incurred; dividends and capital expenditure paid month after declared/incurred; opening cash balance: Rs. 10,000. Management has arranged a bank overdraft of Rs. 40,000.

*Required:*

1. Prepare a cash budget for January–June 2009.

2. Comment on the Managing Director's views in light of the cash budget and suggest possible courses of action.

# Part IV: Flexible Budget

### Key Concepts

A **flexible budget** adjusts for different levels of activity by separating costs into:

| Cost Type | Behaviour |
|---|---|
| **Variable Costs** | Change proportionally with production volume |
| **Fixed Costs** | Remain constant regardless of production volume |
| **Mixed / Semi-variable Costs** | Contain both a fixed and a variable element |

**Variable Cost per Unit** = Budgeted Cost at Base Level ÷ Base Production Level **Flexible Budget Cost** = (Variable Cost per Unit × Actual Units) + Fixed Cost

### Question 1 — Flexible Budget: Materials & Labour *(Partially Solved)*

Base production: 50,000 units; direct materials: $198,000; direct labour: $450,000.

| | **52,500 units** | **60,000 units** | **67,500 units** |
|---|---|---|---|
| Direct Materials (198,000/50,000) × units | 207,900 | | |
| Direct Labour (450,000/50,000) × units | 472,500 | | |

### Question 2 — Flexible Budget: Materials & Labour *(Student Practice)*

Base production: 40,000 units; direct materials: $250,000; direct labour: $350,000.

*Required:* Prepare a flexible budget for production levels of 50,000, 70,000, and 80,000 units.

| | **50,000 units** | **70,000 units** | **80,000 units** |
|---|---|---|---|
| Direct Materials | | | |
| Direct Labour | | | |
| **Total** | | | |

## Question 3 — Flexible Budget: Materials, Labour & Overhead *(Student Practice)*

Base production: 1,000 units; direct materials: $250,000; direct labour: $350,000; factory overhead: $350,000 ($100,000 fixed + $250,000 variable).

*Required:* Prepare a flexible budget for production levels of 50,000, 70,000, and 80,000 units.

| Cost Item | Fixed | Variable/Unit | 50,000 units | 70,000 units | 80,000 units |
|---|---|---|---|---|---|
| Direct Materials | — | | | | |
| Direct Labour | — | | | | |
| Factory Overhead | 100,000 | | | | |
| **Total** | | | | | |

## Question 4 — Flexible Budget: Adding Rent & Utilities *(Student Practice)*

Same base data as Question 3, plus: building rent $50,000 *(fully fixed)*; utility expenses $100,000 ($50,000 fixed + $50,000 variable).

*Required:* Prepare a flexible budget for production levels of 50,000, 70,000, and 80,000 units.

| Cost Item | Fixed | Variable/Unit | 50,000 units | 70,000 units | 80,000 units |
|---|---|---|---|---|---|
| Direct Materials | — | | | | |
| Direct Labour | — | | | | |
| Factory Overhead | 100,000 | | | | |
| Building Rent | 50,000 | — | 50,000 | 50,000 | 50,000 |
| Utility Expenses | 50,000 | | | | |
| **Total** | | | | | |

## Question 5 — Flexible Budget: Adding Repair & Maintenance *(Student Practice)*

Same base data as Question 4, plus: repair & maintenance $200,000 (50% relates to manufacturing; 50% relates to the administrative block — only the manufacturing portion is included in the production flexible budget).

*Required:* Prepare a flexible budget for production levels of 50,000, 70,000, and 80,000 units, incorporating all cost items.

| Cost Item | Fixed | Variable/Unit | 50,000 units | 70,000 units | 80,000 units |
|---|---|---|---|---|---|
| Direct Materials | — | | | | |

| **Cost Item** | **Fixed** | **Variable/Unit** | **50,000 units** | **70,000 units** | **80,000 units** |
|---|---|---|---|---|---|
| Direct Labour | — | | | | |
| Factory Overhead | 100,000 | | | | |
| Building Rent | 50,000 | — | 50,000 | 50,000 | 50,000 |
| Utility Expenses | 50,000 | | | | |
| Repair & Maintenance *(Mfg. portion only)* | | | | | |
| **Total** | | | | | |

# Financial Markets and Institutions

## ◆ 1. Financial Markets (Basic Concept)

**Financial Markets:**
All institutions and procedures for bringing buyers and sellers of financial instruments together.

## ◆ 2. Types of Markets

### ✅ Physical Asset Markets vs Financial Asset Markets

**Physical Asset Markets (also called "tangible" or "real" asset markets):**

- Markets for real products such as:
  - Wheat
  - Automobiles
  - Real estate
  - Computers
  - Machinery

**Financial Asset Markets:**

- Deal with financial instruments such as:
  - Stocks
  - Bonds
  - Notes
  - Mortgages
- Also include **derivative securities**, whose values are derived from other assets
  - Example:
    - Share of Ford stock = pure financial asset
    - Option to buy Ford shares = derivative security

### ✅ Derivatives

**Definition:**
Any financial asset whose value is derived from the value of some other “underlying” asset.

**Example:**

- Bonds backed by subprime mortgages
- Value depends on underlying mortgages

## ✅ Spot Markets vs Futures Markets

**Spot Markets:**

- Assets are bought or sold for **immediate (“on-the-spot”) delivery**

**Futures Markets:**

- Participants agree **today** to buy or sell an asset at a **future date**

## ✅ Money Markets vs Capital Markets

**Money Markets:**

- Markets for **short-term funds (less than one year)**
- Includes:
  - Government and corporate debt securities
  - Securities with less than 1 year to maturity

**Capital Markets:**

- Markets for **long-term securities (more than one year)**
- Includes:
  - Stocks
  - Bonds

## ✅ Primary Markets vs Secondary Markets

**Primary Markets:**

- New securities are issued and sold for the first time

- Companies raise capital

**Secondary Markets:**

- Existing securities are traded among investors
- No new funds go to the company

## ✅ Private Markets vs Public Markets

**Private Markets:**

- Transactions occur directly between two parties
- No public involvement

**Public Markets:**

- Standardized contracts
- Traded on organized exchanges
- Open to public

## ✅ Physical Location Exchanges vs OTC vs Dealer Markets

**Physical Location Exchanges:**

- Formal organizations
- Have physical trading locations
- Conduct auction markets

**Over-the-Counter (OTC) Market:**

- Network of brokers and dealers
- Connected electronically
- Trade unlisted securities

**Dealer Market:**

- Includes facilities for trading securities not listed on exchanges
- Dealers buy and sell from inventory

# ◆ 3. Importance of Financial Markets

Financial markets are essential for a healthy economy because:

1. Provide investment opportunities to savers
2. Provide funds to organizations
3. Help companies raise capital
4. Generate income for the government
5. Attract international investments

# ◆ 4. Financial Institutions

## ✅ Commercial Bank vs Investment Bank

**Commercial Bank:**

- Accept deposits
- Provide loans
- Serve individuals and businesses

**Investment Bank:**

- Underwrites new securities
- Helps businesses obtain financing
- Distributes securities

## ✅ Financial Services Corporation

A firm that offers a wide range of financial services, including:

- Investment banking
- Brokerage operations
- Insurance
- Commercial banking

## ✅ Financial Intermediaries

**Definition:**
Institutions that:

- Accept funds from savers
- Use those funds to provide loans and investments

**Examples:**

- Commercial banks
- Insurance companies
- Pension funds
- Mutual funds

## ✅ Mortgage Banker

- Buys and originates mortgages
- Primarily for resale

## ◆ 5. Investment Vehicles

## ✅ Mutual Funds

- Pool funds from investors
- Invest in diversified securities
- Reduce risk through diversification

## ✅ Money Market Funds

- Type of mutual fund
- Invest in:
    - Short-term, low-risk securities
- Allow investors to write checks

## ✅ Exchange Traded Funds (ETFs)

- Similar to mutual funds
- Traded on stock exchanges

## ✅ Hedge Funds

- Accept money from investors
- Invest in various securities
- Differences:
  - Less regulated
  - Higher risk strategies

## ✅ Key Difference

- Mutual funds & ETFs → regulated
- Hedge funds → largely unregulated

# ◆ 6. Business Forms

## ✅ Sole Proprietorship

- Single owner
- Unlimited liability

## ✅ Partnership

- Two or more owners

**Types:**

- **General Partner:** unlimited liability

- **Limited Partner:** limited liability

## ✅ Corporation

- Separate legal entity
- Features:
  - Limited liability
  - Easy transfer of ownership
  - Unlimited life
  - Can raise large capital

## ✅ Limited Liability Company (LLC)

- Limited liability
- Tax benefits of partnership

# ◆ 7. Key Financial Concepts

## ✅ Depreciation

- Allocation of cost of asset over time

**Types:**

- **Straight-line:** equal expense each year
- **Accelerated:** faster write-off
- **Declining balance:** percentage of book value

## ✅ Cash Dividend

- Distribution of earnings to shareholders

### ✅ Capital Gain (Loss)

- Profit or loss from sale of asset

### ✅ Default

- Failure to meet financial obligations

### ✅ Marketability (Liquidity)

- Ability to sell asset quickly without loss

## ◆ 8. Interest Rate Concepts

### ✅ Maturity

- Time until principal is repaid

### ✅ Term Structure of Interest Rates

- Relationship between interest rate and time to maturity

### ✅ Yield Curve

- Graph showing relationship between yield and maturity

### ✅ Inflation

- Rise in general price level of goods and services

# ◆ 9. Initial Public Offering (IPO)

## ✅ Procedure of IPO

1. Company consults banks
2. Banks analyze market conditions
3. Company decides:
    - Number of shares
    - Price of shares
4. Investors apply by filling forms and depositing amount
5. Minimum purchase usually required
6. If oversubscribed → lottery held
7. Shares allocated and certificates issued

## ◆ Oversubscription vs Undersubscription

**Oversubscription:**

- Applications > shares available
- Lottery system used

**Undersubscription:**

- Applications < shares available
- Remaining shares bought by bankers

# ◆ 10. Stock Exchange Trading

## ✅ Trading Process

1. Open account with broker
2. Broker places order
3. Buyer and seller agree
4. Shares transferred to buyer
5. Money transferred to seller
6. Settlement takes about 2 days
7. Broker charges commission

### ✅ Broker Services

- Buy and sell shares on behalf of client
- Provide advice (if requested)
- Provide recommendations

## ◆ 11. Call Option vs Put Option

### ✅ Call Option

- Right to buy
- Buyer expects price to increase

### ✅ Put Option

- Right to sell
- Seller expects price to decrease

## ◆ 12. Additional Clarifications (From Questions Section)

## ✅ Primary vs Secondary Market (Clarified)

- Primary → new shares issued
- Secondary → trading between investors

## ✅ Spot vs Futures (Clarified)

- Spot → immediate exchange
- Futures → agreement now, delivery later

## ✅ Money vs Capital Market (Clarified)

- Money → short-term cash handling
- Capital → long-term investments